ALLEGRETTI GIOVANNI

Senior C++ Software Engineer

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LinkedIn
Rome (IT)
Portfolio

SUMMARY

Senior C++ Software Engineer with six years of experience in performance-critical development, Qt/QML desktop applications, and computer vision. Strong background in software architecture design and CI/CD integration. Passionate about robotics and embedded systems.

EXPERIENCE

DMA Accurate & Reliable - Software Engineer
Apr 2023 – Present, Turin (IT) - remote

- Lead architectural decisions on the real-time video inspection framework — execution model, threading strategy, and system composition — alongside API design and performance-critical core development.
- Designed the GPU-accelerated inference layer (TensorRT) and scaled it across multiple cards to meet throughput requirements of the most demanding deployments.
- Led the full redesign of the catenary wear measurement product: ground-up Qt/QML UI rewrite, generalization of the analysis from 1 to N wires, and rebuild of the multi-source data synchronization.
- Designed and built a standalone Qt/QML real-time monitoring application around the inspection framework — live camera streams, defect detection results, runtime benchmarks.
- Containerized the core libraries, packaged dependencies into a self-hosted Debian/APT repository, and built multi-platform CI/CD pipelines (Linux GCC + Windows MSVC).

ALTEN (consultant at Leonardo Aircraft) - Software Engineer
Jul 2020 – Mar 2023, Turin (IT)

- Designed and developed HMI applications in C++ and Qt for avionics mission simulation, enabling real-time monitoring of cockpit inputs and injection of simulated failures.
- Developed and maintained a flight and mission simulation model in C++, focusing on reliability, performance, and system integration.

ALTEC - **Aerospace Logistics Technology Engineering Company** - Software Engineer (Internship)
Mar 2019 – Jul 2019, Turin (IT)

- Designed, developed, and tested a C++ software module for planetary exploration robots, computing terrain-adaptive kinematics for a 6-DOF robotic arm. Integrated into an existing ROS/Gazebo framework, with roll/pitch error ≤ 0.10 rad and Z error ≤ 0.10 m on inclined slopes.

PROJECTS AND COLLABORATIONS

ROS Object Grasping — Cognitive Robotics
Università Degli Studi di Salerno 2019, Fisciano (IT)

- Designed and developed a cognitive robotics system for object recognition and manipulation using computer vision and neural networks.

Autonomous Vision Rover — Computer Vision
Università Degli Studi di Salerno 2019, Fisciano (IT)

- Developed an autonomous rover system for navigation in a controlled environment.

STM32 Differential-Drive Rover - Embedded Systems
Università Degli Studi di Salerno 2018, Fisciano (IT)

- Designed and implemented a custom PID closed-loop controller and firmware for both manual and autonomous (line-following) driving modes on an STM32F401RETx.

LANGUAGES

Italian
Native ●●●●●

English
Advanced ●●●○○

TECHNICAL SKILLS

Core
C++ **Qt/QML** **Computer Vision** **Multithreading** **Software Architecture** **Design Patterns**

Tools & CI/CD
Git **GitLab** **CI/CD** **Docker** **CMake** **GTest** **Valgrind** **Kestra** **DebianPackaging** **vcpkg** **gdb** **Intel VTune** **GitHub**

Libraries & Frameworks
OpenCV **Boost** **ZMQ** **ROS** **TensorFlow** **TensorRT** **Gazebo**

Platforms
Linux **Windows** **Embedded (ESP32, STM32)** **Nvidia Jetson**

Other Languages
Python **C** **MATLAB** **Bash**

Methodologies
Agile/Scrum

PERSONAL TRAITS

Problem Solving **Teamwork** **Critical Thinking** **Attention to Detail** **Continuous Learning**

EDUCATION

Facial Expression Recognition - Computer Vision

Master thesis Erasmus Program - University of Groningen 2019, Groningen (NL)

- Automatic Facial Expression Recognition in unconstrained ('in-the-wild') video, using temporal modelling (CNN+LSTM, C3D) and multimodal video+audio fusion.

Master's Degree in Computer Engineering **110**/110

Università Degli Studi di Salerno 2020, Fisciano (IT)

Bachelor's Degree in Computer Engineering **95**/110

Università Degli Studi di Salerno 2017, Fisciano (IT)